

Question 1

NOT COMPLETE

Case Study Screen 1 of 6

The following 6 questions refer to this clinical scenario: NURS 1017 Unit 5 Case Study 6.

Phase 1 — Emergency Department, 0305.

➤ The nurse reviews the Nurses' Notes. **Highlight** the findings that indicate the client is at risk for airway compromise or impaired gas exchange.

Nurses' Notes

0305:

Client is tearful and repeatedly asking about his dog. Oriented to person only; unsure of the time or place. Reports headache and nausea. Voice is hoarse and he has a frequent cough producing black-tinged sputum. Nasal hairs and eyebrows singed; soot around the nose and mouth. Face reddened and dry without blisters. Expiratory wheezes heard bilaterally. Anterior chest and abdomen bright red and moist with intact blisters of varying size; blanches with pressure; rates chest pain 9/10. Entire right arm, circumferentially from shoulder to wrist, is waxy white and tan, dry, leathery, and nonblanchable; no pain when touched. Wearing a wedding ring on the right hand and a metal watch on the right wrist. Right radial pulse 2+; capillary refill 3 seconds.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

Question 1

COMPLETE

Case Study Screen 1 of 6

The following 6 questions refer to this clinical scenario: NURS 1017 Unit 5 Case Study 6.

Phase 1 — Emergency Department, 0305.

The nurse reviews the Nurses' Notes. **Highlight** the findings that indicate the client is at risk for airway compromise or impaired gas exchange.

Nurses' Notes

0305: Client is tearful and repeatedly asking about his dog. Oriented to person only; unsure of the time or place. Reports headache and nausea. Voice is hoarse and he has a frequent cough producing black-tinged sputum. Nasal hairs and eyebrows singed; soot around the nose and mouth. Face reddened and dry without blisters. Expiratory wheezes heard bilaterally. Anterior chest and abdomen bright red and moist with intact blisters of varying size; blanches with pressure; rates chest pain 9/10. Entire right arm, circumferentially from shoulder to wrist, is waxy white and tan, dry, leathery, and nonblanchable; no pain when touched. Wearing a wedding ring on the right hand and a metal watch on the right wrist. Right radial pulse 2+; capillary refill 3 seconds.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

CORRECT

Score: 5 / 5 points

Rationale:

New confusion (oriented to person only) after smoke exposure suggests cerebral hypoxia from carbon monoxide (CO) and/or impaired gas exchange. Hoarseness signals thermal injury and swelling of the larynx and upper airway; edema can progress rapidly to obstruction, so this is a red-flag cue. Black-tinged (carbonaceous) sputum shows that smoke was inhaled below the vocal cords. Singed nasal hairs and eyebrows with soot around the nose and mouth are classic external cues of smoke and heat inhalation, especially after confinement in an enclosed, smoke-filled space. Bilateral expiratory wheezes indicate bronchospasm and lower-airway irritation from inhaled smoke and chemicals. Being tearful and asking about his dog is an expected emotional response that needs support later, but is not a cue of airway or gas-exchange risk. Chest pain 9/10 comes from the partial-thickness (second-degree) burn to the anterior trunk, which is very painful because nerve endings are exposed; it needs treatment, but it is not an airway cue. The circumferential full-thickness right arm burn is a circulation concern (risk for compartment syndrome), not an airway concern in this client -- a circumferential full-thickness burn of the *chest* could restrict breathing, but the arm cannot. The ring and watch on the burned arm must be removed before swelling develops (a circulation concern), but do not indicate airway risk.

Question 2

NOT COMPLETE

Case Study Screen 2 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 1 — Emergency Department, 0305.

- For each assessment finding, click to specify whether it is consistent with **upper airway (inhalation) injury**, **carbon monoxide poisoning**, or **hypovolemia related to the burn injury**. Each finding may support more than one condition.

Finding / Action	Upper airway (inhalation) injury	Carbon monoxide poisoning	Hypovolemia related to burn
Hoarse voice	☐	☐	☐
Singed nasal hairs	☐	☐	☐
Headache and nausea	☐	☐	☐
Oriented to person only	☐	☐	☐
SpO2 98% despite new confusion	☐	☐	☐
HR 122/min with BP 104/66 mm Hg	☐	☐	☐
Hematocrit 0.53 L/L (53%)	☐	☐	☐

Note: Each column must have at least 1 response option selected.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

Question 2

COMPLETE

Case Study Screen 2 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 1 — Emergency Department, 0305.

- For each assessment finding, click to specify whether it is consistent with **upper airway (inhalation) injury**, **carbon monoxide poisoning**, or **hypovolemia related to the burn injury**. Each finding may support more than one condition.

Finding / Action	Upper airway (inhalation) injury	Carbon monoxide poisoning	Hypovolemia related to burn
Hoarse voice	☒	☐	☐
Singed nasal hairs	☒	☐	☐
Headache and nausea	☐	☒	☐
Oriented to person only	☒	☒	☐
SpO2 98% despite new confusion	☐	☒	☐
HR 122/min with BP 104/66 mm Hg	☐	☐	☒
Hematocrit 0.53 L/L (53%)	☐	☐	☒

Note: Each column must have at least 1 response option selected.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

CORRECT

Score: 8 / 8 points

Rationale:

Hoarse voice and singed nasal hairs are direct evidence of heat and smoke near the airway (upper airway/inhalation injury). Headache and nausea are early, nonspecific symptoms of CO toxicity, since CO binds hemoglobin far more tightly than oxygen, reducing oxygen delivery to tissues. Confusion (oriented to person only) can come from hypoxemia caused by airway and lung injury, and from tissue hypoxia caused by CO -- both apply. A normal-appearing SpO2 of 98% despite new confusion is falsely reassuring: standard pulse oximeters cannot tell carboxyhemoglobin from oxyhemoglobin, so the COHb level on the ABG (ordered, pending) is needed to detect CO poisoning. Burns over 20% TBSA trigger a systemic inflammatory response with fluid shifts that can lead to hypovolemic shock; tachycardia (HR 122) with a narrowing blood pressure (104/66) is compensation for that fluid loss. A hematocrit of 53% reflects hemoconcentration: plasma leaks out of the vessels while red cells stay in, raising the hematocrit as intravascular volume falls.

Question 3

NOT COMPLETE

Case Study Screen 3 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 1 — Emergency Department, 0305.

➤ Complete the following sentences by choosing from the lists of options.

The client is at highest risk for

Select...

 as evidenced by

Select...

. Once this is addressed, the nurse should next focus on the client's risk for

Select...

 as evidenced by

Select...

.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

Question 3

COMPLETE

Case Study Screen 3 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 1 — Emergency Department, 0305.

➤ Complete the following sentences by choosing from the lists of options.

The client is at highest risk for

airway obstruction from upper airway edema

 as evidenced by

the hoarse voice and black-tinged sputum

 . Once this is addressed, the nurse should next focus on the client's risk for

hypovolemic shock

 as evidenced by

HR 122/min, BP 104/66 mm Hg, and hematocrit 0.53 L/L

 .

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

CORRECT

Score: 2 / 2 points

Rationale:

The nurse prioritizes with ABCDE, and for burns, ABC is followed by fluid resuscitation. After smoke inhalation, impaired gas exchange takes priority over impaired skin integrity. Upper-airway edema can worsen over hours, so early recognition (the hoarse voice and carbonaceous sputum) allows a controlled intubation before the airway is lost entirely. Circulation follows airway and breathing: with about 27% TBSA of partial- and full-thickness burns, large fluid shifts are expected, and IV access and fluid resuscitation are top priorities after the airway, evidenced by the tachycardia, low-normal blood pressure, and hemoconcentration (elevated hematocrit). Wound infection develops over days, not within this timeframe. Hypothermia is a real risk but is addressed after airway, breathing, and circulation. Compartment syndrome of the arm threatens a limb and needs frequent checks, but a threat to the airway threatens life.

Question 4

NOT COMPLETE

Case Study Screen 4 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

Provider Orders

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 2 — Emergency Department, 0340.

- The nurse plans the client's fluid resuscitation. Use the Nurses' Notes (Phase 1), the Rule of Nines, and the provider's orders. Complete the following sentences by choosing from the lists of options.

The percentage of TBSA to include in the fluid resuscitation calculation is

Select...▼

. The total volume of Ringer's lactate prescribed for the first 24 hours is

Select...▼

. Half of this volume,

Select...▼

, is to be infused over the first 8 hours, timed from

Select...▼

. The nurse will titrate the infusion as prescribed to maintain a urine output of at least

Select...▼

.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

Question 4

COMPLETE

Case Study Screen 4 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

Provider Orders

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 2 — Emergency Department, 0340.

- The nurse plans the client's fluid resuscitation. Use the Nurses' Notes (Phase 1), the Rule of Nines, and the provider's orders. Complete the following sentences by choosing from the lists of options.

The percentage of TBSA to include in the fluid resuscitation calculation is

27%

 . The total volume of Ringer's lactate prescribed for the first 24 hours is

8,640 mL

 . Half of this volume,

4,320 mL

 , is to be infused over the first 8 hours, timed from

0215 (time of injury)

 . The nurse will titrate the infusion as prescribed to maintain a urine output of at least

40 mL/h

 .

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

CORRECT

Score: 5 / 5 points

Rationale:

By the Rule of Nines: anterior trunk (chest + abdomen) = 18%; the entire right arm, anterior 4.5% + posterior 4.5% = 9%. Total 27%. The face is a first-degree (superficial) burn -- red, dry, no blisters -- and is **not** counted in resuscitation calculations. Using the Parkland formula, 4 mL × 80 kg × 27 = 8,640 mL over 24 hours. Half of that, 4,320 mL, is given in the first 8 hours; the remaining 4,320 mL is given over the next 16 hours. The 8-hour clock starts at the burn, not at arrival -- because 45+ minutes have already passed since 0215, the hourly rate must make up for lost time. The adult urine-output target is about 0.5 mL/kg/h: 0.5 × 80 kg = 40 mL/h. Urine output is the key guide for titration; output well above 1 mL/kg/h (for example, 100–160 mL/h) suggests over-resuscitation and risks edema-related complications.

Question 5

NOT COMPLETE

Case Study Screen 5 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

Provider Orders

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 2 — Emergency Department, 0340.

- For each nursing action, click to specify whether it is **indicated**, **contraindicated**, or **nonessential** for the client at this time.

Finding / Action	Indicated	Contraindicated	Nonessential
Notify the provider immediately about the right-hand findings	☐	☐	☐
Remove the wedding ring and watch from the right hand and wrist	☐	☐	☐
Elevate the right arm on pillows above the level of the heart	☐	☐	☐
Apply a snug elastic compression wrap to the right arm to limit swelling	☐	☐	☐
Continue 100% oxygen via non-rebreather mask until the airway is secured	☐	☐	☐
Administer the prescribed morphine intramuscularly into the unburned thigh	☐	☐	☐
Apply ice packs to the blistered chest to relieve pain	☐	☐	☐
Offer small sips of an electrolyte drink to replace fluid losses	☐	☐	☐
Teach active range-of-motion exercises for the right hand	☐	☐	☐
Cover the burns with clean, dry dressings or sheets and keep the room warm	☐	☐	☐

Note: Each row must have only 1 response option selected.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

Question 5

COMPLETE

Case Study Screen 5 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

Provider Orders

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 2 — Emergency Department, 0340.

- For each nursing action, click to specify whether it is **indicated**, **contraindicated**, or **nonessential** for the client at this time.

Finding / Action	Indicated	Contraindicated	Nonessential
Notify the provider immediately about the right-hand findings	☒	☐	☐
Remove the wedding ring and watch from the right hand and wrist	☒	☐	☐
Elevate the right arm on pillows above the level of the heart	☒	☐	☐
Apply a snug elastic compression wrap to the right arm to limit swelling	☐	☒	☐
Continue 100% oxygen via non-rebreather mask until the airway is secured	☒	☐	☐
Administer the prescribed morphine intramuscularly into the unburned thigh	☐	☒	☐
Apply ice packs to the blistered chest to relieve pain	☐	☒	☐
Offer small sips of an electrolyte drink to replace fluid losses	☐	☒	☐
Teach active range-of-motion exercises for the right hand	☐	☐	☒
Cover the burns with clean, dry dressings or sheets and keep the room warm	☒	☐	☐

Note: Each row must have only 1 response option selected.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

CORRECT

Score: 10 / 10 points

Rationale:

Cool, pale fingers, delayed capillary refill, a weakening pulse, paresthesia, and a tense forearm under a circumferential full-thickness burn signal impending compartment syndrome; the provider must be notified immediately, and the expected treatment is an escharotomy (incision through the inelastic eschar). Jewelry and clothing near the burn should be removed, since once edema develops, a ring or watch acts as a tourniquet. Elevating the arm above heart level reduces edema formation. Anything constricting (a compression wrap) worsens the compartment pressure that is already compromising perfusion. A carboxyhemoglobin of 19% confirms CO poisoning; high-flow oxygen shortens the half-life of carboxyhemoglobin and continues until the airway is secured and the COHb falls. During burn shock, peripheral and muscle perfusion is poor, so IM absorption is unreliable -- little early relief, then possible delayed, excessive absorption once perfusion returns; the client's orders specify IV morphine. Ice causes vasoconstriction that can deepen the injury and, over a 27% TBSA burn, promotes hypothermia; cooling with room-temperature water or saline is used instead. The client is NPO with an NG tube ordered: intubation is imminent and large burns commonly cause paralytic ileus, so resuscitation and any fluids are

Question 6

NOT COMPLETE

Case Study Screen 6 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

Provider Orders

Progress Notes

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 3 — Regional Burn Centre, Post-injury Day 5.

- For each finding on post-injury day 5, click to specify whether it indicates that the plan of care has been **effective** or **not effective**.

Finding / Action	Effective	Not effective
Urine output averaging 55 mL/h over the past 24 hours	☐	☐
Right hand warm, capillary refill 2 seconds, right radial pulse 2+	☐	☐
SpO2 96% on O2 2 L/min nasal cannula; lungs clear; speaks in full sentences	☐	☐
Pain rated 3/10 during a dressing change after IV morphine 30 minutes beforehand	☐	☐
Green-yellow, foul-smelling drainage from the chest wound; redness extending 2 cm beyond the wound edge	☐	☐
Temperature 38.9 °C; WBC 17.4 × 10 ⁹ /L	☐	☐

Note: Each row must have only 1 response option selected.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

Question 6

COMPLETE

Case Study Screen 6 of 6

The nurse is caring for a 46-year-old male client brought to the ED after a house fire, with suspected inhalation injury.

History and Physical

Vital Signs

Laboratory Results

Nurses' Notes

Provider Orders

Progress Notes

History and Physical:46-year-old male brought in by EMS after a house fire that began at approximately 0215. Firefighters found him in a smoke-filled bedroom. Medical history: hypertension (lisinopril 10 mg daily). No known drug allergies. Tetanus immunization status unknown. Stated weight 80 kg.

Phase 3 — Regional Burn Centre, Post-injury Day 5.

- For each finding on post-injury day 5, click to specify whether it indicates that the plan of care has been **effective** or **not effective**.

Finding / Action	Effective	Not effective
Urine output averaging 55 mL/h over the past 24 hours	☒	☐
Right hand warm, capillary refill 2 seconds, right radial pulse 2+	☒	☐
SpO2 96% on O2 2 L/min nasal cannula; lungs clear; speaks in full sentences	☒	☐
Pain rated 3/10 during a dressing change after IV morphine 30 minutes beforehand	☒	☐
Green-yellow, foul-smelling drainage from the chest wound; redness extending 2 cm beyond the wound edge	☐	☒
Temperature 38.9 °C; WBC 17.4 × 10 ⁹ /L	☐	☒

Note: Each row must have only 1 response option selected.

Submit

Adapted from OpenStax Medical-Surgical Nursing (2024), Chapter 14 — Integumentary System (CC BY-NC-SA 4.0). This derivative work is licensed CC BY-NC-SA 4.0.

CORRECT

Score: 6 / 6 points

Rationale:

Urine output of 55 mL/h is above the 40 mL/h (0.5 mL/kg/h) target, showing adequate renal perfusion after resuscitation. A warm hand with brisk capillary refill and a palpable pulse matches the expected short-term outcome of no signs of compartment syndrome: no swelling and no decreased pulses in the affected area. SpO2 96%, clear lungs, and speaking in full sentences show that gas exchange and airway patency have been restored after extubation. Pain 3/10 during a dressing change reflects giving analgesia at least 30 minutes before burn care, with pain that is tolerable before, during, and after the dressing change. Yellow, foul-smelling discharge with increased erythema are signs of burn-wound infection; the nurse notifies the provider (a wound culture and antimicrobials are likely). The short-term goal is a WBC within normal range; fever with leukocytosis and a purulent wound suggest infection and possible sepsis, so the provider is notified and this is escalated promptly.